2. Extending an existing ontology gives you the benefits of adopting an existing standard, but with the ability to extend it to support your specific needs where necessary. The main drawback of this approach is the need to manage your extensions to the standard coupled with reduced support from other tools and systems.
3. Creating a custom ontology will enable you to create concepts that align perfectly with your domain at the expense of not having a shared vocabulary with other systems which may limit your interoperability.

For the digital twin of my home, I evaluated the Digital Buildings Project from Google (https://github.com/google/digitalbuildings) and the ontology defined by Microsoft in DTDL for smart buildings, which is based on the RealEstateCore ontology (https://www.realestatecore.io/) —an ontology for property owners. I chose the RealEstateCore-based DTDL as it is better suited to my home whereas the Digital Buildings ontology is better suited to large, modern, commercial buildings. The major concepts and relationships in the DTDL ontology are shown in figure 5.5. Where the model does not support concepts or relationships that I would like to model (for example a relationship between an asset and documents), I will extend it.